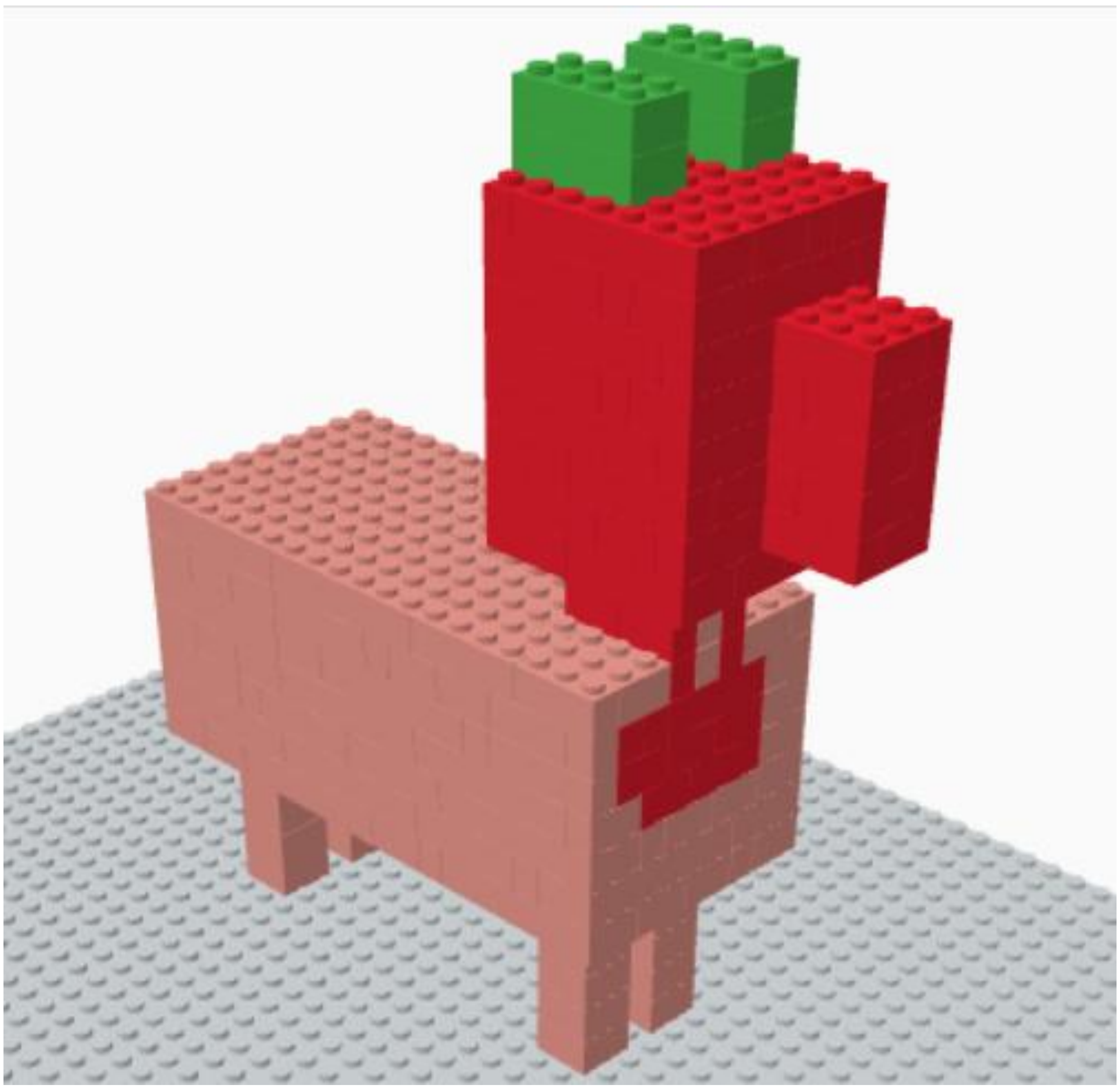
My Preliminary Mechanical Design Report

Quadruped Robot Dog



Introduction:

To fulfill the practical requirements of this assignment, I have successfully designed and built the preliminary 3D mechanical structure of a quadruped robot dog using Tinkercad. Based on the course objectives, my design focuses on the fundamental mechanical principles that enable stable standing and locomotion. Below is the comprehensive engineering analysis of my design.

1-Body Shape and Structure:

For the robot's main body, I chose a box-style, blocky chassis. I found that using this rectangular geometry is the most practical approach because it gives me enough internal space to safely house all the electronic components, wires, and the battery. It also keeps the structure rigid, so it doesn't twist or flex under mechanical stress.

2-Locomotion Design and Degrees of Freedom (DoF):

- Leg Placement: I placed the four legs symmetrically at the outer corners of the body frame. This layout creates a wide, rectangular base of support, which maximizes stability.

- Degrees of Freedom: I decided to configure the robot with 8 Degrees of Freedom (8 DoF) in total.

    1. Each leg has 2 DoF (two active joints operated by motors): one at the hip (Hip Pitch) to swing the leg forward and backward, and one at the knee (Knee Pitch) to bend and retract the foot.

    2. I selected this 2 DoF setup per leg to minimize overall weight and reduce control complexity while ensuring the feet can easily clear the ground during the swing phase.

3-Actuator Selection:

For the joints, I selected the TowerPro MG996R Servo Motors instead of smaller options like the SG90.

- Engineering Justification: Since the legs must directly support and lift the robot's body, they face constant load. The MG996R servos feature metal gears, which provide high wear resistance against cyclic stress, and they deliver a high stall torque of up to 11kg.cm at 6V, making them ideal for holding joints locked underweight.

4. Preliminary Dynamic Calculations & Stability

- I conducted a mathematical check to verify if the selected MG996R servos can support the robot in the worst-case mechanical leverage (when a leg is fully extended horizontally at a 90degrees angle.

1. Assumptions & Inputs:

   - Estimated total robot mass(m) = 1.2kg
   - Total Weight Force (W= m * g) = 1.2 * 9.81 =11.8 N
   - Worst-case dynamic phase: The total weight is temporarily supported by only 2 legs (W (per leg)) = 11.8 / 2 = 5.9N
   - Maximum effective leg length acting as a lever arm (r) = 8 cm= 0.08 m

2. Torque Equation:

$\tau = F * r * \sin(\theta)$

($\sin(90) = 1$, Since $\theta = 90$ for maximum leverage)

$\tau = 5.9N * 0.08m = 0.472N.m$

3. Safety Factor Evaluation:

   - 4.8 kg.cm Converting 0.472 N.m to standard servo units yields approximately
   - Calculating the Safety Factor (SF):
     - SF = τ stall / τ required =11kg.cm / 4.8 kg.cm = 2.3
   - Since SF > 2, the selected motors are highly capable of handling both static holding loads and dynamic walking forces without overheating.

5. Stability and Center of Mass (COM):

- Center of Mass (COM): I plan to distribute the heavy components primarily the LiPo battery centrally within the chassis. This keeps the COM exactly at the geometric center of the robot.
- Static Balance Condition: For the robot to remain balanced without falling, the vertical projection of its COM must always lie within the Support Polygon formed by the feet touching the floor.

6. Proposed Locomotion: Static Crawl Gait

I chose to implement a Static Crawl Gait (Creep Gait) for this project.

1. Locomotion Sequence: The robot moves using a 3+1sequence, meaning it lifts and steps only one leg at a time while keeping the other three feet firmly on the ground.
2. My Reasoning: By moving one leg at a time, the remaining three legs always form a stable Support Triangle. If the programmed sequence ensures the COM projection stays inside this triangle before the next foot lift, the robot maintains static balance at every fraction of a second. This eliminates the need for expensive inertial measurement sensors (IMUs) and keeps the programming manageable as a foundational prototype.

7. Anticipated Mechanical Issues & My Solutions

1. Joint Backlash (Wobbling): Small clearances between the servo gear and the leg attachment can cause structural shaking. My Solution*:* I will use tight mechanical tolerances and 3D-printed support brackets to distribute load away from the servo horn.
2. Impact Shock at Foot Strike: Repetitive ground impacts can damage the servo gear teeth over time.My solution*:* I will attach high-friction rubber pads to the tips of the feet to act as shock absorbers.
3. Battery Shifting: If the battery slides during walking, the COM shifts out of the support triangle, causing the robot to fall. My Solution: I am designing a dedicated, snug locking compartment in the center of the Tinkercad chassis to keep the battery completely fixed.

Conclusion:

In summary, the 3D prototype I built in Tinkercad serves as a solid structural framework that successfully balances simplicity with fundamental mechanical principles. It fulfills all requirements, ensuring that the physical robot will be stable, structurally sound, and ready for walking tests.